Aditya Kumar Singh

Software Development Engineer in Test (SDET) · Performance Engineering · Cloud-Native Quality Engineering · ISTQB CT-FL Certified

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PROFESSIONAL SUMMARY

SDET with 3.5+ years building test automation, performance engineering, and observability for enterprise insurance platforms at Duck Creek Technologies (via NashTech). Owns test framework design, performance benchmarking, and incident response end-to-end — from test strategy through production postmortems. ISTQB CT-FL certified.

PROFESSIONAL EXPERIENCE

Automation Test Engineer, Level 2 — Promoted Jul 2024

Jul 2024 – Present

NTG Solutions LLP · NashTech Ltd, UK · Delhi NCR, India

NashTech (Nash Squared Group) is a UK-headquartered global technology services firm with operations across Asia, Europe, and North America.

Client: Duck Creek Technologies, Destination Architecture Foundation (Platform, Billing, Claims, AMI)

Duck Creek Technologies is a leading enterprise software platform for Property and Casualty insurance carriers, serving clients globally.

- Designed and owns a layered TypeScript/Jest test framework with a client abstraction layer for the User Admin service, enabling the team to add new REST or gRPC test coverage without touching shared infrastructure code.
- Extended the framework to dual-protocol coverage — 26 REST methods and 22 gRPC methods across 9 service stubs (Protocol Buffers) — spanning 6 functional domains in both positive and negative flows.
- Implemented Faker.js-powered test data factories with Azure AD token acquisition built into the framework; integrated BrowserStack for cloud-based cross-browser execution across environments.
- Evaluated 5 performance testing tools (K6, JMeter, Gatling, Artillery, Locust) and issued a formal K6 adoption recommendation that the team accepted; designed a 4-phase performance testing strategy and executed baseline and peak capacity tests for Unified IAM to establish repeatable scaling benchmarks.
- Designed Grafana dashboard architecture for performance observability, integrating K6 metrics with AKS pod data, MongoDB Atlas connection pools, Kafka consumer lag, and Envoy Gateway queue depth.
- Deployed k6-operator on AKS to distribute load across 10 parallel runner pods, enabling repeatable 45-minute load profiles across User Query and Access Management Query microservices without script changes for every run.
- Statically routed REST and gRPC scenarios to dedicated pod groups using execution-segment splitting, with VU counts and ramp timings driven entirely by environment variables for repeatable, config-only test runs.
- Built a full performance telemetry pipeline (k6 → OpenTelemetry → in-cluster OTel Collector → Prometheus → Grafana Cloud), giving the team real-time per-endpoint p95/p99 visibility during load test runs instead of post-run log digging.
- Instrumented 31 custom latency metrics and 12 error counters across REST and gRPC endpoints, breaking down failures by type (timeouts, 5xx, deadline-exceeded, unavailable) to speed up root-cause triage during test runs.
- Built a single Grafana dashboard consolidating load test results and AKS infrastructure health, so engineers could correlate performance regressions with pod and node resource saturation without switching tools.
  - Dashboard covers 59 panels across VUs, REST/gRPC throughput, latency percentiles, error rates, check pass rates, and node-level CPU, memory, disk, and network metrics.
- Automated pod log capture for load test runs, removing a manual bottleneck caused by developers having no direct kubectl access to the fleet cluster.
  - Built the log capture workflow to authenticate via Azure OIDC, auto-discover all 10 runner pods through the ArgoCD resource-tree API, and archive logs every 2 minutes as a 30-day CI artifact — fully hands-off from test trigger to log availability.
- Owned the end-to-end K6 test deployment pipeline: TypeScript sources bundled via esbuild, baked into Kubernetes ConfigMaps through Kustomize, promoted across environments via Kargo, and synced to the AKS fleet cluster by ArgoCD — enabling zero-touch test deployment on every push to main.

- Designed active-passive DR strategies for platform services, defining service-level RTO/RPO expectations and failover classifications (auto vs manual); executed HA verification scenarios via failure injection across multiple services.
- Authored the cross-regional Kafka failover runbook covering configuration validation, regional service correctness, and API sanity verification.
- Traced a SEV-1 production outage (16-hour impact across 3 carrier teams, 70+ failed requests) to a configuration drift between routing and API management layers, using Grafana, Splunk, and Azure KQL.
- Turned the SEV-1 investigation into reusable Splunk query playbooks so on-call engineers could diagnose similar drift issues without repeating the manual trace; co-authored this and other Post-Incident Reviews across platform services.
- Shipped 3 monitoring improvements directly from incident postmortems: HTTP error alerting to catch infra failures before customer impact, SLA monitoring to surface delivery breaches proactively, and a CI stabilization window that eliminated false failures from post-deployment automation running too early.
- Enforced quality gates across 4 product workstreams serving live insurance carriers: mandatory unit test coverage for happy-path and failure scenarios, Snyk security scans maintaining zero critical vulnerabilities, and regression suite validation on every feature branch before production merge.
- Defined and documented the end-to-end deployment lifecycle across five environments (Playground to Production), embedding quality gates, approval checkpoints, and post-deployment verification standards that made releases consistent, auditable, and repeatable across the team.
- Authored consumer-facing API integration documentation for platform notification and Core services, providing self-serve request shapes, required headers, and copy-ready examples that reduced integration support overhead for dependent engineering teams.
- Mentored L1 consultants and interns through code reviews, dual-protocol framework design, and ISTQB-aligned testing methodology.

Automation Test Engineer, Level 1 — Promoted Aug 2023 | Aug 2023 – Jun 2024

NTG Solutions LLP · NashTech Ltd, UK · Delhi NCR, India

Client: Duck Creek Technologies, Destination Architecture Foundation (Platform, Billing, Claims, AMI)

- Built and executed automated test suites for frontend (Cypress) and backend (.NET/XUnit, C#) components across Billing, Claims, and Platform services; authored BDD scenarios using Cucumber and SpecFlow (Gherkin).
- Conducted API testing and response contract validation via Postman and Rest Assured against Azure Function App and Azure Container App endpoints over HTTP/HTTPS.
- Performed functional, regression, smoke, cross-browser, compatibility, system integration, and responsive design testing across sprint deliverables; managed test cycles and defect lifecycle in Azure Test Plans.
- Carried out database testing against SQL Server and Azure Cosmos DB to validate data integrity and transactional correctness across service boundaries.
- Contributed to event-driven architecture testing, validating Kafka-backed asynchronous service workflows and CQRS write-to-read consistency.
- Collaborated closely with developers and product owners during sprint planning and refinement to identify edge cases early, reducing defect leakage and ensuring test coverage reflected real-world usage rather than just stated acceptance criteria.

Software Engineer Intern | Feb 2023 – Jul 2023

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- Completed NashTech's structured Automation and Manual Testing onboarding programme, gaining proficiency in Selenium, Java, and core QA methodologies aligned with ISTQB standards.
- Authored functional and regression test scripts; maintained test documentation and defect reports within CI/CD workflows using Jenkins and Git.
- Participated in Agile sprint ceremonies and contributed to sprint deliverables under senior engineer mentorship.

KEY PROJECT ENGAGEMENT

Destination Architecture Foundation: Platform Services, Billing, Claims and AMI

Duck Creek Technologies (global enterprise P&C insurance software platform) · via NashTech Ltd, UK · Aug 2023 – Present

Platform Services: Centralised notification and user-preference management; User Admin service (REST + gRPC, CQRS, Kafka,

AMI: Subscription-based integrations with CoreLogic, LexisNexis, NCCI, Verisk and Hyland OnBase.

multi-region DR).

Billing: End-to-end policy billing covering billable items, payment collection, disbursements, commissions, and write-offs.

Claims: Web API bridge between the Claims front-end platform and Duck Creek backend services.

Stack: TypeScript · Jest · gRPC / Protocol Buffers · .NET Core 7.0 · C# · XUnit · SpecFlow · Cypress · Kafka (Confluent) · MongoDB Atlas · Azure AKS · Azure Function Apps · Azure Container Apps · Azure Cosmos DB · Azure APIM · Azure Application Gateway · SQL Server · K6 · Grafana · Splunk · BrowserStack · GitHub · Azure DevOps

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, C#, Java, Python

Test Frameworks: Jest, Cypress, Selenium, XUnit, TestNG, JUnit, NightwatchJS, Robot Framework

API and Contract Testing: Postman, Rest Assured, Supertest, Axios, gRPC / Protocol Buffers

BDD: Cucumber, SpecFlow (Gherkin), Behaviour-Driven Development

Performance Testing: K6, k6-operator, Apache JMeter, Gatling

Observability: Grafana (dashboards and alerting), Splunk (log analysis), OpenTelemetry, KQL (Azure Diagnostics)

Cloud - Azure: AKS, Kubernetes, Function Apps, Container Apps, Cosmos DB, APIM, App Gateway, AD, Traffic Manager

Cloud - AWS: Amazon AWS (general exposure)

Event Streaming: Apache Kafka (Confluent), CQRS pattern, Event-Driven Architecture

CI/CD and GitOps: GitHub Actions, Azure DevOps, Jenkins, ArgoCD, Kargo, Kustomize, Git, GitHub, Snyk, NuGet

Databases: SQL Server, MongoDB Atlas, Azure Cosmos DB, MySQL

Test Data and Auth: Faker.js, UUID, Azure AD / @azure/identity (OAuth2)

OS and Protocols: Windows, Linux (Ubuntu), HTTP/HTTPS, gRPC, Protocol Buffers

Methodologies: Agile, Scrum, Kanban, BDD, TDD, CQRS, Clean Code, Incident Response (PIR/RCA)

CERTIFICATIONS

ISTQB Certified Tester – Foundation Level (CT-FL) · ISTQB · Jul 2023 · No Expiry

ISTQB Agile Testing and Automation Testing Foundation – Knowledge Certified · ISTQB

ISTQB® Certified Tester AI Testing (CT-AI) v2.0 · ISTQB

Reactive Architecture Fundamentals (LB0101EN) · IBM · Aug 2023

Building Scalable Systems (LB0107EN) · IBM · Aug 2023

Innovative India Coding Championship, Certificate of Appreciation · Jul 2022

COMMUNITY AND KNOWLEDGE SHARING

AI Assisted Test Failure Analysis NashLearn · NashTech

youtu.be/3NxoCRXgCFk

Applying AI-assisted techniques to accelerate root-cause analysis of test failures in enterprise automation pipelines.

Smart Test Data: Faker vs AI Knolx · NashTech

youtube.com/watch?v=0H5A6l1fAHI

Practical comparison of Faker.js against AI-generated data strategies in automated testing pipelines.

Testing Event-Driven Architecture Knolx · NashTech

youtube.com/watch?v=TRmYxFd9RG4

Strategies for validating Kafka-backed, asynchronous, and CQRS systems drawn from real-world enterprise experience.

Clean Code in Test Automation Knolx · NashTech

youtube.com/watch?v=7X9lnn7kDNY

Applying clean code principles to automation frameworks: layered architecture, abstraction, maintainability, and readability in test suites.

Technical Writing LinkedIn Articles

linkedin.com/in/its-aks/recent-activity/articles

Published articles on quality engineering, test automation, cloud-native testing practices, and engineering culture.

EDUCATION

Master of Computer Applications (MCA)

GL Bajaj Institute of Technology and Management · Dr. A.P.J. Abdul Kalam Technical University (AKTU), Greater Noida · 2021 – 2023

Bachelor of Commerce, Honours (B.Com Hons.)

Swami Shraddhanand College · University of Delhi · 2018 – 2021

LANGUAGES

English: Full Professional Proficiency **Hindi:** Native